Describe a wave

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify amplitude, wavelength, frequency and period.

SUCCESS CRITERIA - I can...

1. Measure amplitude from equilibrium.
2. Measure wavelength between matching points.
3. Connect frequency and period.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A sample warms by 3°C after a certain energy input. If absorbed energy becomes 5 times as large with the same mass and material and no phase change, find the new rise.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: f = cycles/Δt; T = 1/f.
f is frequency (Hz); T is period (s). λ is crest-to-crest wavelength; amplitude is maximum displacement.

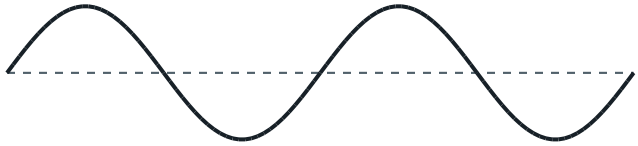
Worked reasoning: A wave makes 6 cycles in 3 s: f=2 Hz and period T=0.5 s. Amplitude measures maximum displacement, not the spacing of crests.

Your turn: Draw one spatial wave and label amplitude and wavelength. Separately mark one period on a displacement-time graph.

Marzano check - circle: ① ② ③

03 Look and explain

Label amplitude and one wavelength on the wave diagram. Explain why crest-to-trough height is not the amplitude.



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Build the solution

A point completes 6 cycles in 5 s. Find frequency. If one cycle takes .5 s, state the period.

Marzano check - circle: ① ② ③

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05 Use the idea

Does increasing amplitude necessarily increase frequency? Use the definitions to explain.

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg sample has c=600 J/(kg·°C) and warms by 8°C without changing phase. Find Q.

Marzano check - circle: 1 2 3

07 Exit check

A wave has period 2 s. How many cycles occur in 6 s? Find frequency.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Draw two waves with equal wavelength but different amplitudes. Explain which feature changes and which does not.

Marzano check - circle: 1 2 3